

CAI6734 Milestone Report

Testing in Imagination:
World-Model Planning for Language Agents
in Sequential Clinical Diagnosis

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Abstract

We extend LA-CDM from retrospective, model-free test selection to world-model-based *testing in imagination*: an agent that can evaluate counterfactual diagnostic actions via a learned transition model $p_\phi(o_{t+1} | h_t, a_t)$. This milestone summarizes problem framing, baseline archaeology, HiPerGator-compliant model choices, Stage 1 synthetic environment scaffolding, and next deliverables. Full MIMIC Table 1 reproduction is deferred until PhysioNet access and CUDA jobs are available; we log honest gaps rather than invent metrics.

1 Project goal

Language agents for clinical decision-making (CDM) typically either assume all patient information is available up front or interact with a retrospective chart where many tests were never ordered. LA-CDM trains a LoRA-adapted language agent with GRPO to iteratively request tests and diagnose under cost pressure, but it remains *model-free*: when a test is unavailable in the chart, the environment returns “unavailable,” which is **not** a counterfactual of what that test would have shown.

Our research question is whether a learned world model $p_\phi(o_{t+1} | h_t, a_t)$ enables planners to *test in imagination*—and whether a flawed world model can make planning *worse* than model-free baselines (a publishable negative result). We compare model-free LA-CDM-style agents, world-model planners, and an oracle world model (available only in synthetic or imputed settings).

Team & repository. Private GitHub: <https://github.com/scalinity/testing-in-imagination> (usernames: scalinity, EdwinASalcedo, haresendez). Advisor: Xuefeng Liu.

2 Baseline: LA-CDM (reproduce carefully)

Paper: Bani-Harouni et al., *Language Agents for Hypothesis-driven Clinical Decision Making with Reinforcement Learning*, ICLR 2026 (arXiv:2506.13474).

Code: <https://github.com/dharouni/LA-CDM>.

Upstream method (shared LLM weights):

- **Hypothesis agent** $\mathcal{H} : p_j \mapsto \{h_j, c_j\}$ (diagnosis hypothesis + confidence 0–10).
- **Decision agent** $\mathcal{D}$ (ReAct) that orders a test or commits a diagnosis.
- Cyclic training: action GRPO $\rightarrow$ hypothesis SFT $\rightarrow$ confidence GRPO; LoRA on an ~ 7 –8B instruct backbone.

Headline paper claims (Table 1) we will only fill from logged runs: mean Acc 81.3, Micro F1 84.1, Macro F1 81.3, avg test cost \$1295.61. We treat GitHub clone $\neq$ reproduced until config, seed, and metric artifacts are logged (`REPRODUCTION.md`).

2.1 Archaeology findings (honest gaps)

ID	Finding
D-1	Paper narrative uses <code>incorrect_reward = -1.0</code> ; shipped Hydra config uses 0.0. Reproduce shipped config first.
D-2	Physical-exam cost present in config ($\approx$ \$300) but missing from Appendix cost table; twelve costs sum to 1.0.
D-3	Paper HPI summarizer = Mixtral-8x7B; upstream <code>prepare_data.py</code> defaults to Qwen2.5-7B.
D-4	No canonical 80/10/10 split seed $\Rightarrow$ expect Table 1 mismatch; quantify with ≥ 3 seeds.
F-1–F-3	Unavailable tests share error flags with bad actions; unavailable requests are <i>free</i> and skip hypothesis update; weak pressure to avoid them.
B-1	Full stack needs CUDA/vLLM; local Mac cannot produce Table 1. Prefer HiPerGator.
D-5	CoC backbone swap (below).

Thesis implication. The baseline environment makes unavailability cheap and uninformative. A world model changes the economics so every planned action yields a predicted observation—including counterfactuals the chart never recorded.

3 HiPerGator constraint and backbone choice

UF countries-of-concern guidance prohibits PRC-origin LLMs on HiPerGator (Qwen, DeepSeek, Yi, InternLM, and derivatives), even via LoRA or “re-releases.” Upstream LA-CDM defaults to Qwen/Qwen2.5-7B-Instruct, so bit-for-bit weight match to their Table 1 setup is not feasible on the cluster.

Our default (logged as D-5).

- **Policy + HPI summarizer:** `nvidia/Llama-3.1-Nemotron-Nano-8B-v1`
Meta Llama 3.1 8B Instruct **US base** + NVIDIA post-training (reasoning / tool-calling). UFIT RC confirmed NVIDIA models are allowed when the base is US.

- **Fallback ablation:** meta-llama/Llama-3.1-8B-Instruct (same family, no Nemotron post-train).
- **World model p_ϕ :** tiny bag-of-words / small classifier first—not another frontier LLM.

This is a documented reproduction deviation. We do not claim Table 1 bit-for-bit weight match. Hydra/`prepare_data` defaults that still point at Qwen must be overridden before any GRPO or vLLM job; HF revision / RC mirror path will be pinned when cluster access is confirmed.

4 Stage 1 progress (implementation)

4.1 Documentation & research process

- Standing constitution (`ONBOARDING.md`), claims ledger (`REPRODUCTION.md`), paper notes, env/tuple specs, open questions.
- Formalized POMDP-style interface: history h_t , actions $\mathcal{A}_{\text{test}} \cup \mathcal{A}_{\text{diag}}$, observation o_{t+1} , objective $p_\phi(o_{t+1} \mid h_t, a_t)$.

4.2 Synthetic CDM environment (scaffolded)

Under `extension/envs/synthetic_diagnosis/` (no MIMIC/PHI; pure Python + pytest):

- Latent disease y over {appendicitis, cholecystitis, diverticulitis, pancreatitis}.
- Twelve LA-CDM-aligned test actions with template observations $g_a(y, \epsilon)$.
- Always-available v0 (no UNAVAILABLE) for clean world-model tuple collection; `oracle_observe` for counterfactual probes.
- JSONL tuple export CLI (`collect.py`): random / expert stub policies; diagnosis steps filtered out of WM training tuples.
- Unit tests under `extension/tests`.

4.3 What we deliberately deferred

- Full PhysioNet / MIMIC-IV-Ext-CDM prep and Table 1 GRPO runs (awaiting credentials + HiPerGator GPU).
- Treating AnyJev/Jev (typed calibrated decisions) as a new research question—useful as optional decision plumbing later, but would dilute Stage 1’s world-model claim.

5 Evaluation plan (next milestones)

1. **Phase E:** unit tests for LA-CDM parsers, rewards, unavailable branch, stub metrics (Mac-OK).
2. **Tiny p_ϕ :** bag-of-words / small classifier on synthetic JSONL; plug into a simple imagination planner.

3. **Comparisons:** model-free vs WM-planner vs oracle WM; include a deliberately degraded ϕ that can hurt planning.
4. **Partial baseline (B):** zero-shot / smoke eval on HiPerGator with Nemotron Nano once data + CUDA are ready; full train only under logged seed/config.
5. Pin Nemotron HF revision / RC mirror; override all Qwen Hydra defaults before any cluster job.

6 Division of labor (working)

- **Daniel Escalante:** research scaffolding, reproduction ledger, CoC backbone decision & docs, synthetic env design coordination, repo stewardship (`scalinity/testing-in-imagination`).
- **Hugo Resendez / Edwin Salcedo:** collaborator onboarding via shared GitHub; next: Phase E tests, Hydra overrides, and HiPerGator job wiring as access is confirmed.

Roles will be refined as MIMIC access and GPU allocation land; this report reflects work completed through Week 5.

7 Risks and mitigations

- **CoC / cluster:** mitigated by Nemotron Nano (US base) + RC confirmation; keep stock Llama 3.1 8B as ablation.
- **Table 1 mismatch:** expected (D-4, D-5); document deviations; never invent numbers.
- **Scope creep:** keep AnyJev optional; Stage 1 claim stays world-model planning on synthetic CDM then MIMIC.
- **PHI:** no PhysioNet extracts in the public-facing workflow; synthetic-first path avoids early PHI risk.

References

1. D. Bani-Harouni, C. Pellegrini, E. Özsoy, M. Keicher, and N. Navab. Language Agents for Hypothesis-driven Clinical Decision Making with Reinforcement Learning. In *ICLR*, 2026. arXiv:2506.13474. Code: <https://github.com/dharouni/LA-CDM>.
2. Private course repository: <https://github.com/scalinity/testing-in-imagination>.